

Lecture 8: Doing the Right Thing

HUM 4441: Engineering Ethics

Reaz Hassan Joarder, Samnun Azfar, and Yousuf Ibne Kamal Niloy

Junior Lecturer

{reazhassan, samnunazfar, yousufniloy}@iut-dhaka.edu

Department of CSE

Islamic University of Technology

The Engineer's Dilemma

Engineers must design to meet NHTSA's binding regulations. IIHS results are not binding at all — but they say something real about how a car behaves in the kind of crash that is actually most common: an offset, not a full frontal, collision.

Decide: An engineer at an automaker learns that a model comfortably meets NHTSA's standard but would score IIHS's lowest rating. What should they recommend to management, and does the cost of the fix change your answer?

Legal compliance is a floor, not a ceiling, when better safety information already exists.

Fleddermann, Ch. 8, p. 154

Avoiding Impediments to Ethical Behavior

Many engineering ethics problems have an obvious right answer, yet engineers do not always choose it. Three excuses come up again and again.

"It's not my problem"

Someone else's mistake, someone else's decision.

"If I don't, someone else will"

Refusing does not stop the technology from being built.

"I can't foresee everything"

No one can predict every consequence of a new design.

Each excuse contains some truth — which is exactly what makes it easy to hide behind.

Fleddermann, Ch. 8, p. 153

Testing the First Two Excuses

“It’s not my problem”

- Lawsuits and redesigns after an unsafe product raise prices for everyone who buys it.
- Injuries raise health-insurance premiums for everyone.
- Contract fraud is made up by taxpayers.

It really is everyone’s problem.

“If I don’t, someone else will”

- Often true — you are rarely the only one working on a given idea.
- But the question is not whether it gets built; it is whether you want to be first to build something harmful.

Being first to notoriety is not a distinction worth having.

Fleddermann, Ch. 8, p. 153

Building In Foresight

"I can't foresee everything" is also true — but engineering is a creative process, and foresight can be built into that process rather than left to chance.

- Design around problems identified in advance, and work with regulators before a new technology ships.
- Treat ethical considerations as part of the design process, not an afterthought.
- Accept that some designs simply should not be built.

If genuine foresight still misses a consequence, that is different from negligence — the right response is to work afterward to make things right, as LeMessurier did.

Fleddermann, Ch. 8, pp. 153–155